

Lab : Understanding and Calculating File Hashes Using Windows Server and Kali Linux

Objective: To apply the process of calculating, transferring, and verifying file hashes using Windows Server 2022 and Kali virtual machine to enhance the understanding of file integrity and security practices

Tools required: Windows Server 2022 and Kali Virtual Machine

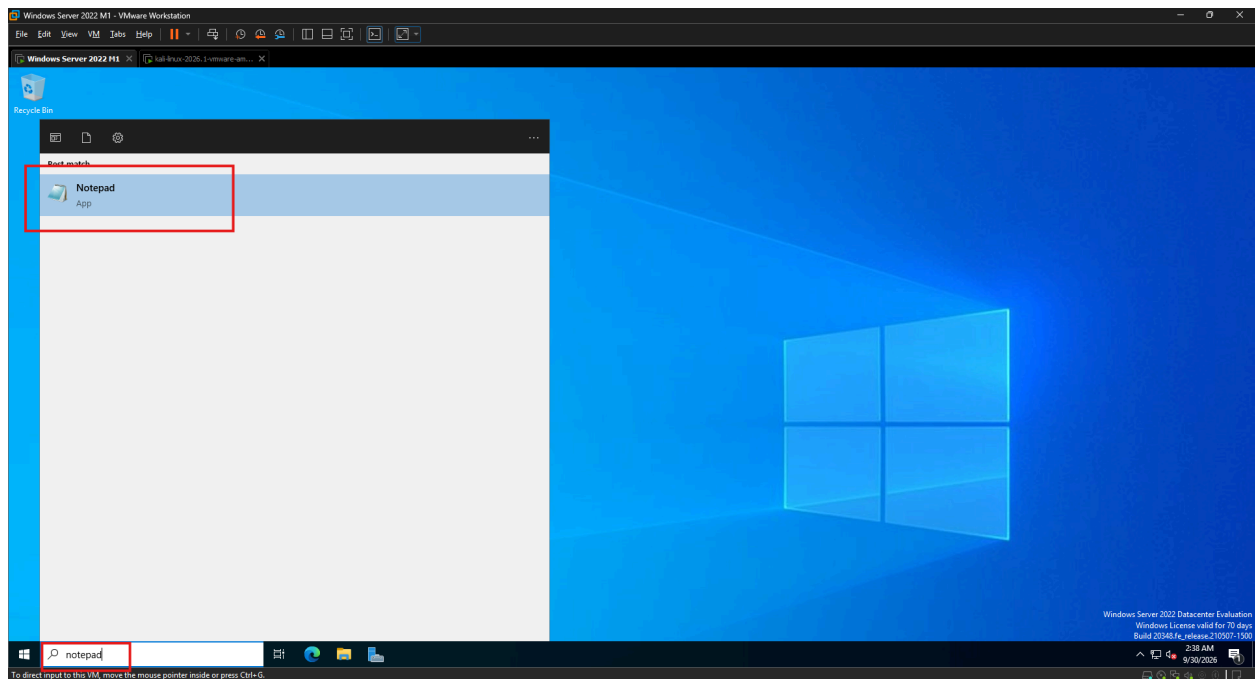
Prerequisites: None

Steps to be followed:

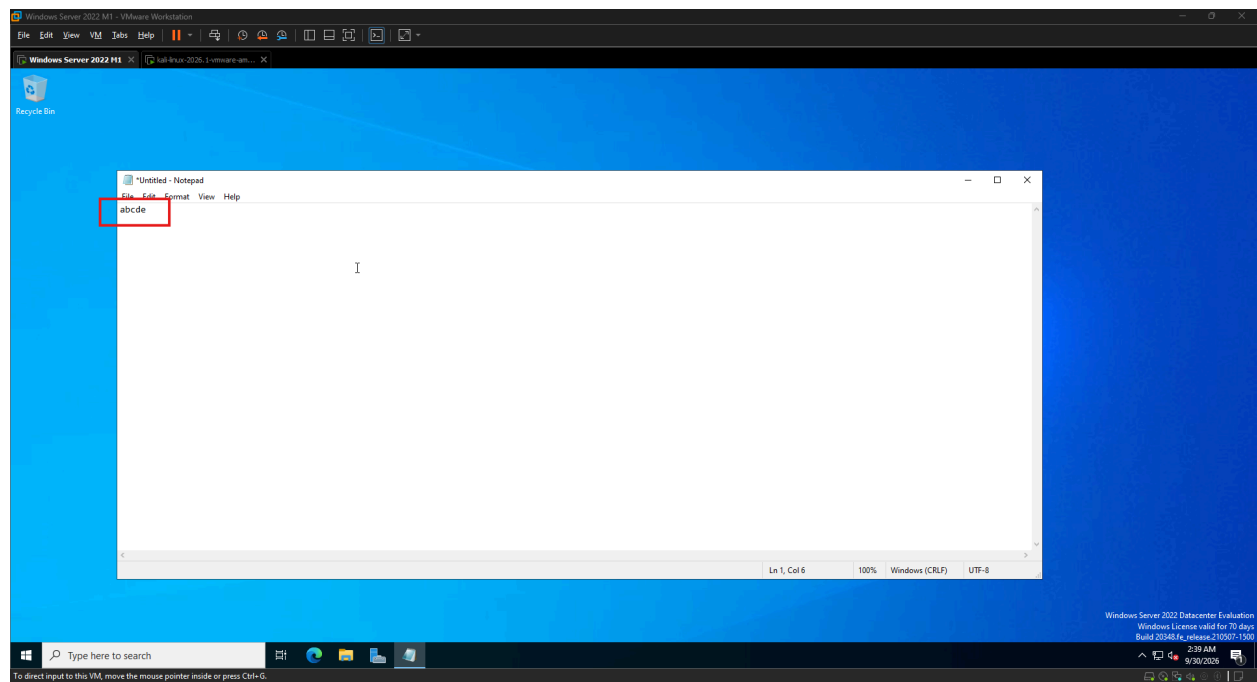
1. Create the file and calculate its hash
2. Transfer and verify the file on the Kali virtual machine

Step 1: Create the file and calculate its hash

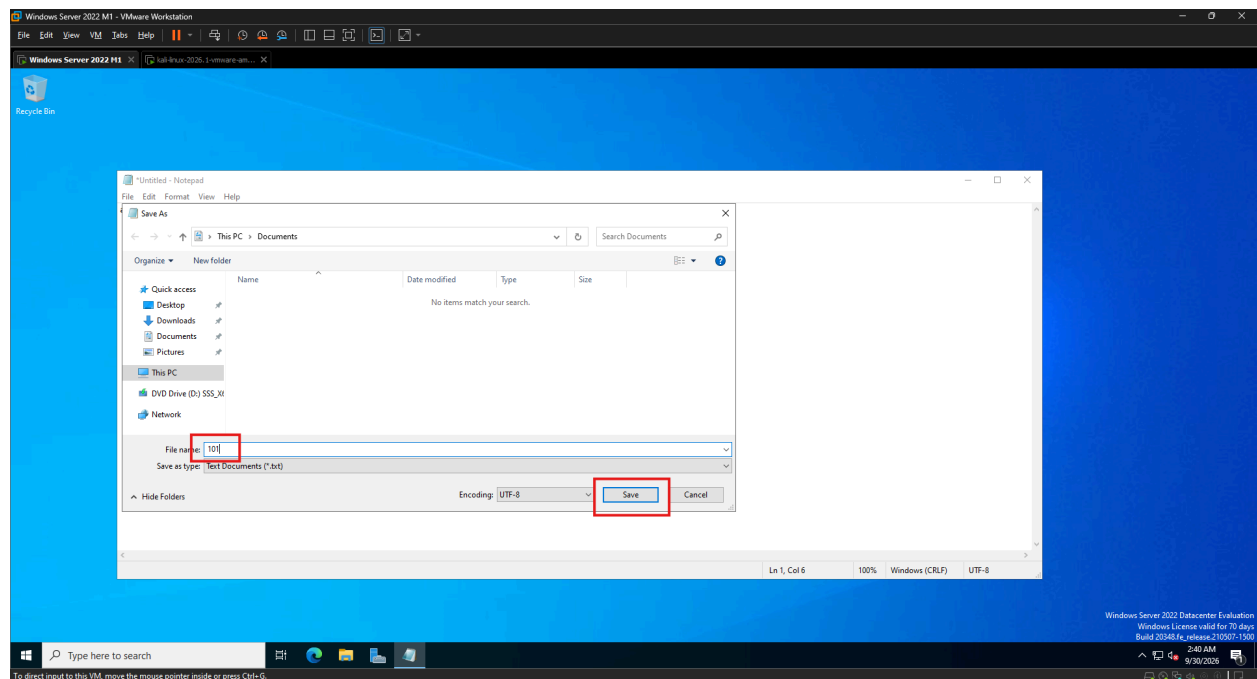
1.1 Open Notepad in Windows Server 2022 VM in VMware



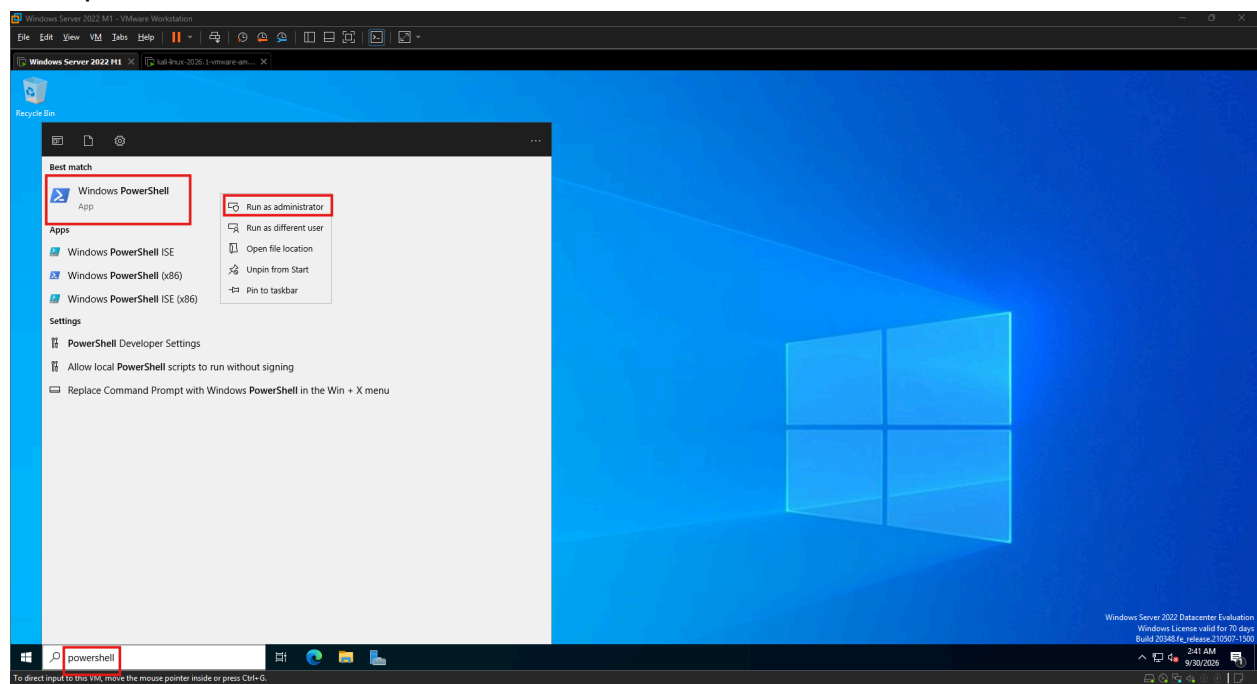
1.2 Type the following text inside the Notepad: abcde



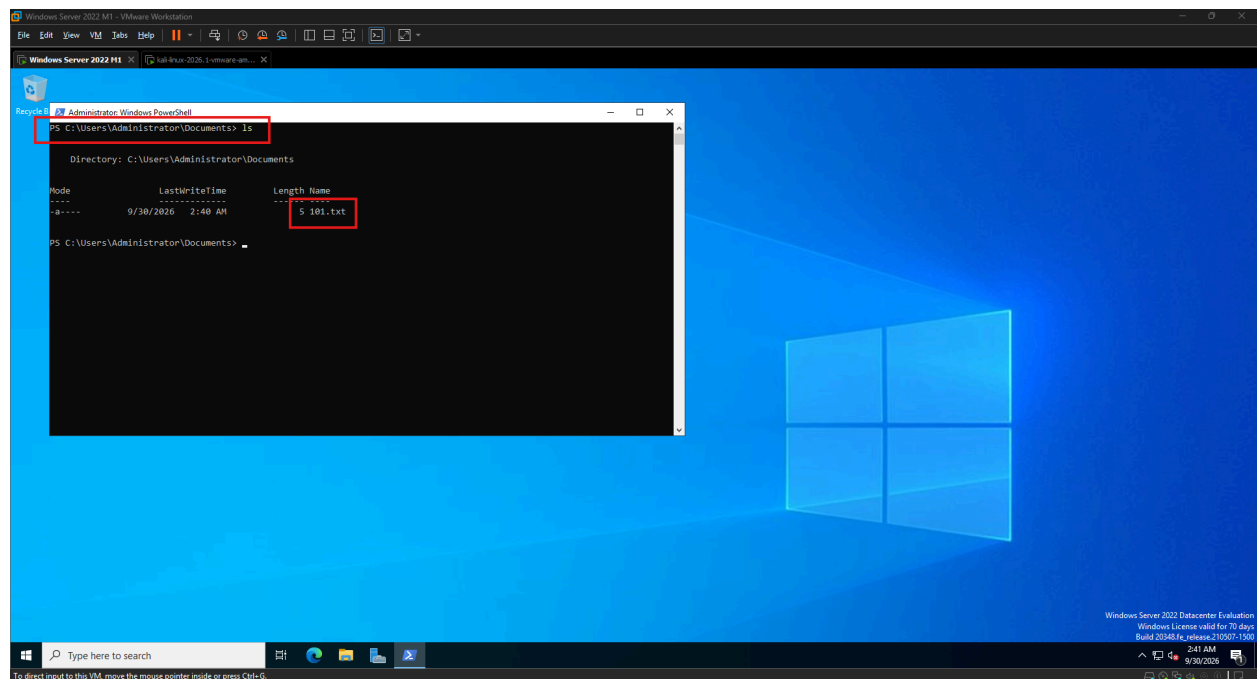
1.3 Name the file 101.txt and save it



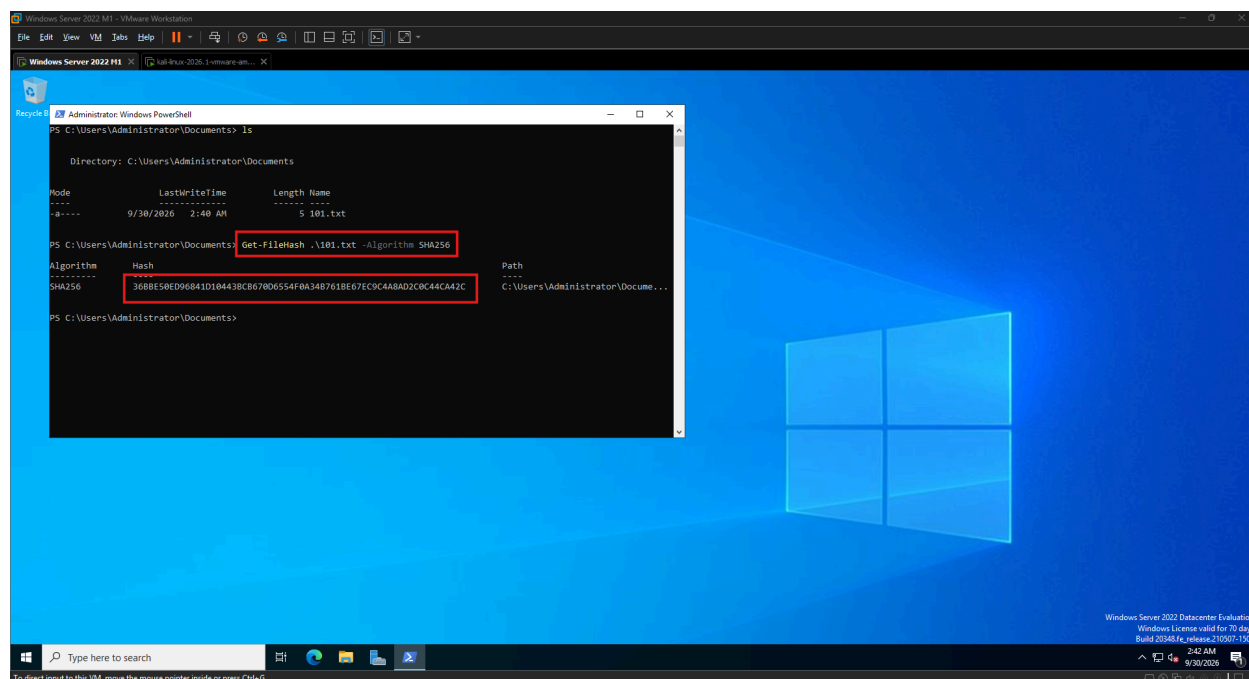
1.4 Open PowerShell in Run as Administrator mode



1.5 Run the following command: *ls* to view the created file

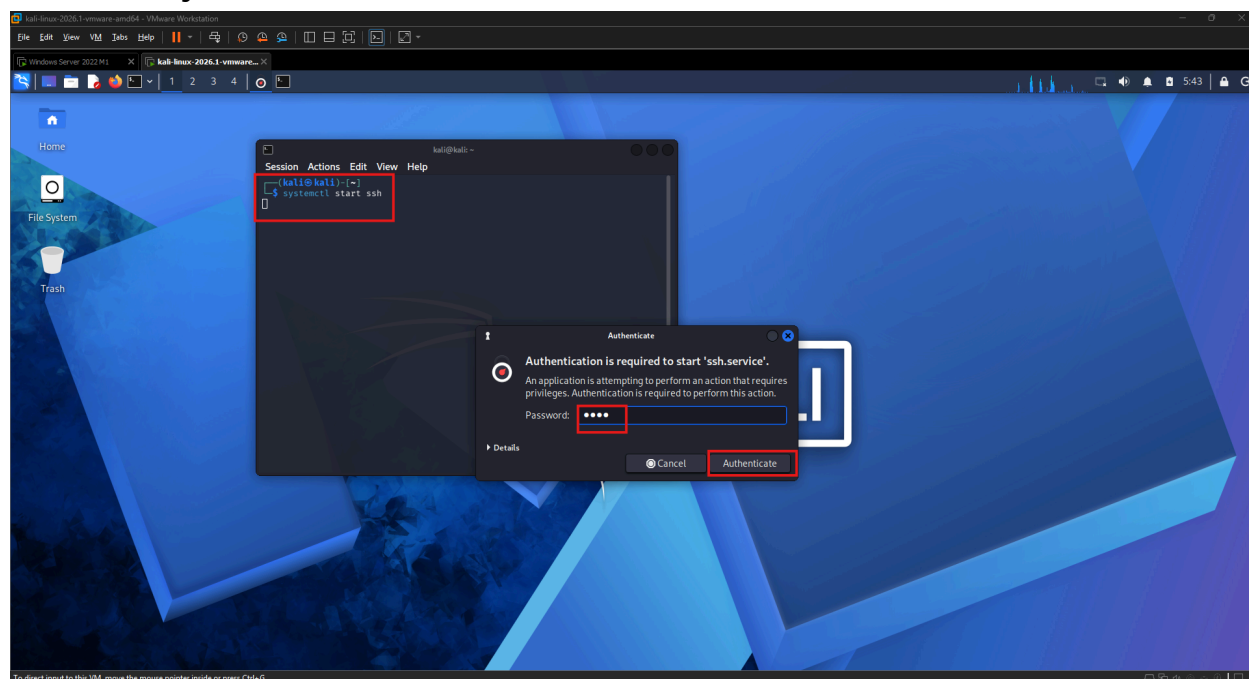


1.6 Run the following command: ***Get-FileHash 101.txt -Algorithm SHA256*** to find the hash of the created file



Step 2: Transfer and verify the file on the Kali virtual machine

2.1 Start Kali Virtual Machine and open the Terminal Emulator and type the following command: ***systemctl start ssh***

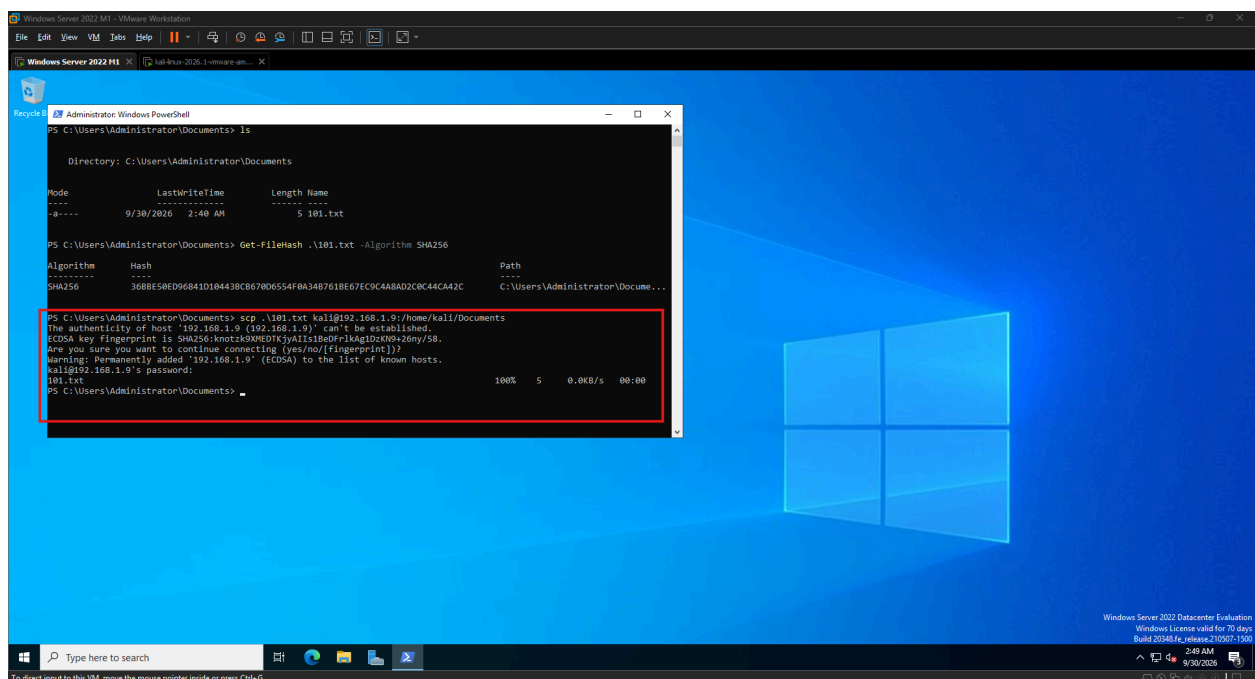
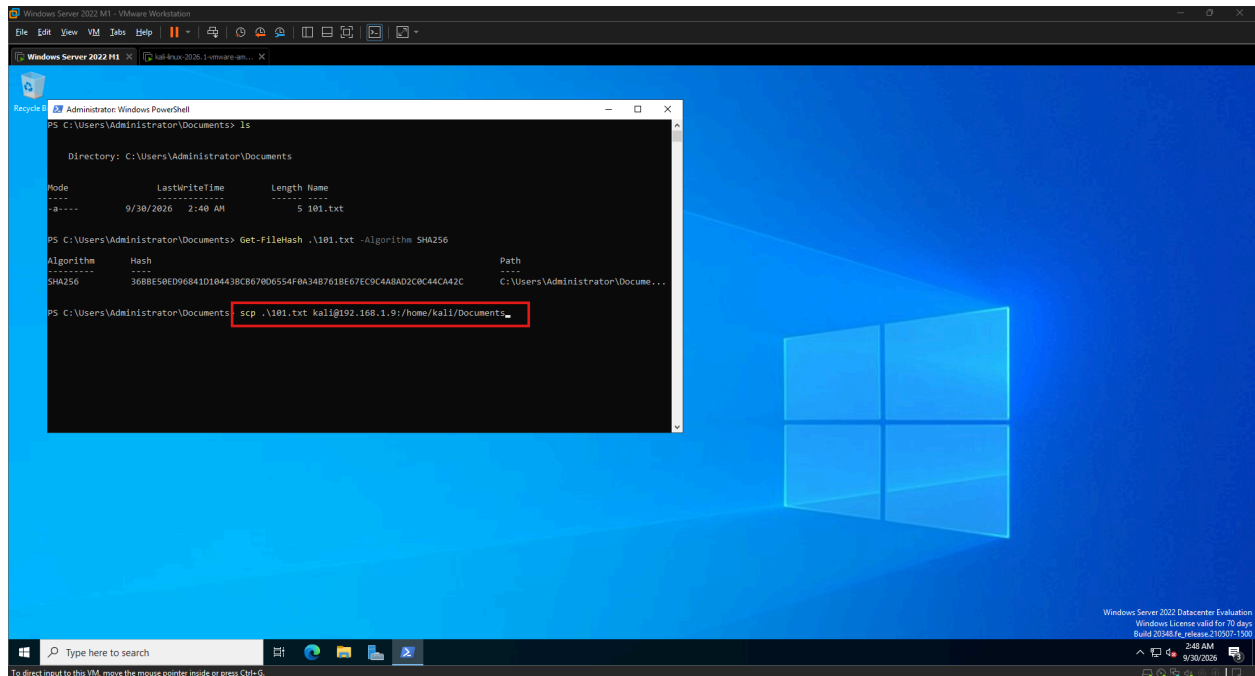


2.2 Type the following command: **ip a** to view the IP address of your system

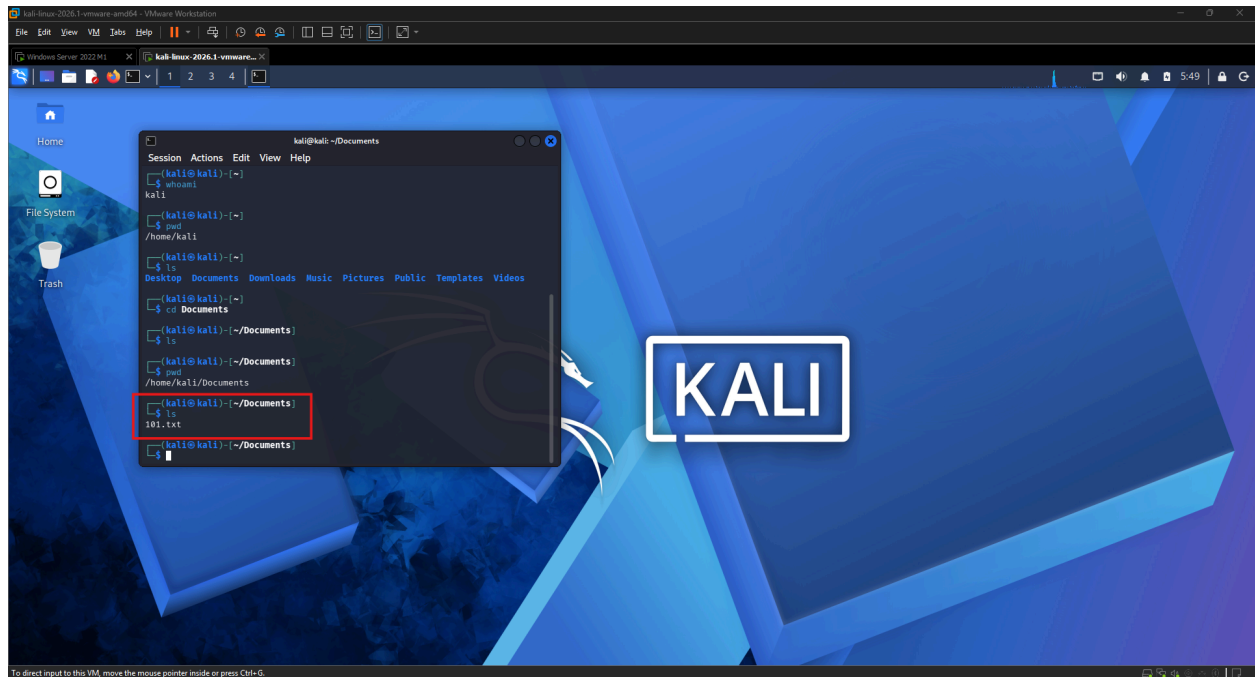


2.3 Navigate back to PowerShell in Windows Server 2022 and enter the following command to transfer the 101.txt file in the Kali Linux machine:

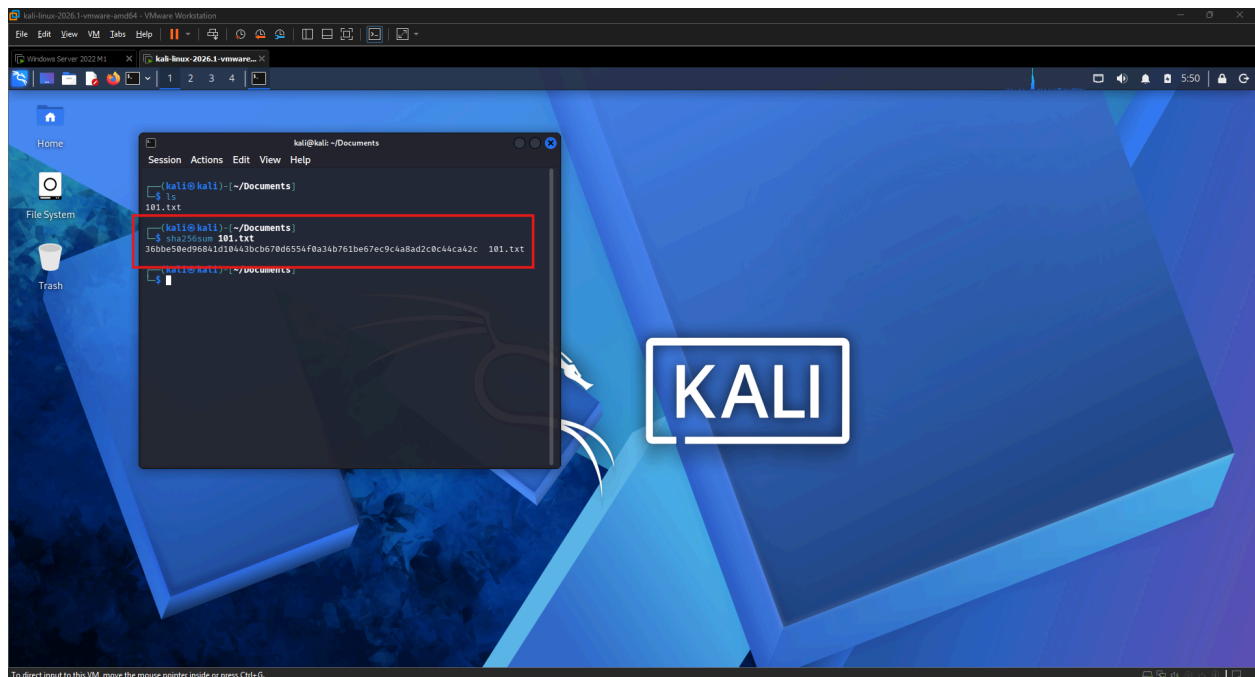
scp 101.txt labuser@192.168.1.9:/home/kali/Documents



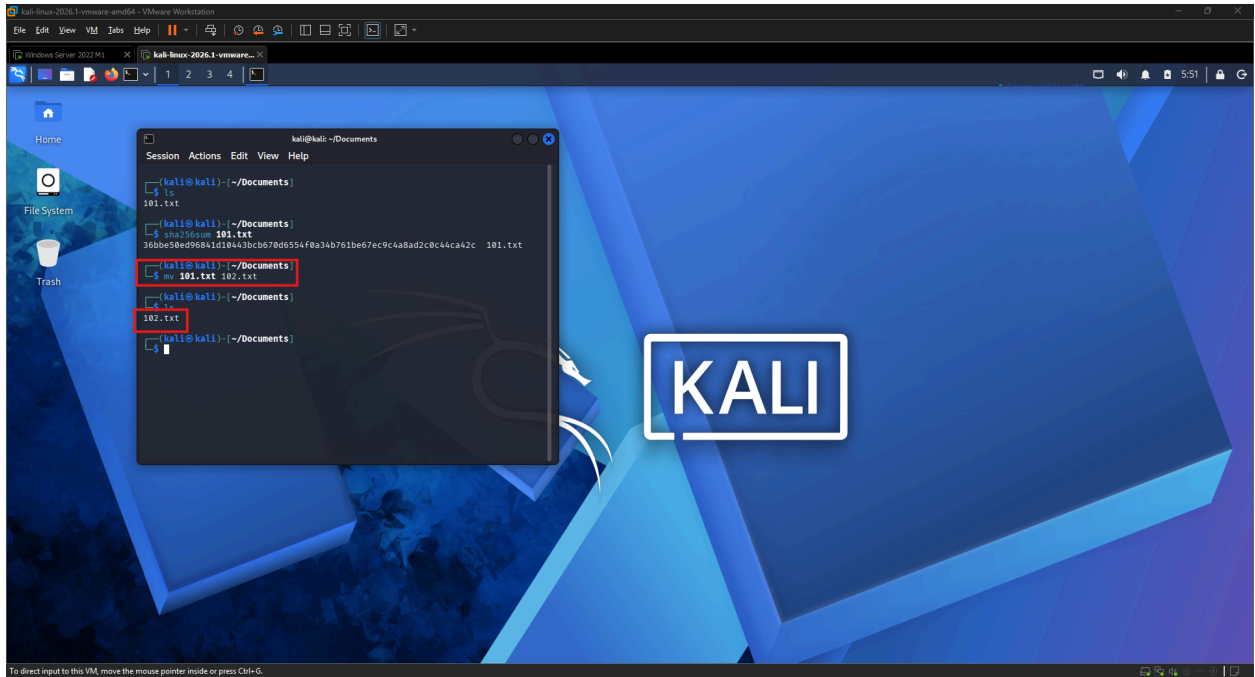
2.4 Navigate back to Kali virtual machine and enter the following command in the terminal: ***ls*** to view the 101.txt



2.5 Type the following command to verify the hash of the transferred file :
sha256sum 101.txt



2.6 Type the following command: ***mv 101.txt 102.txt*** to rename the file from 101.txt to 102.txt

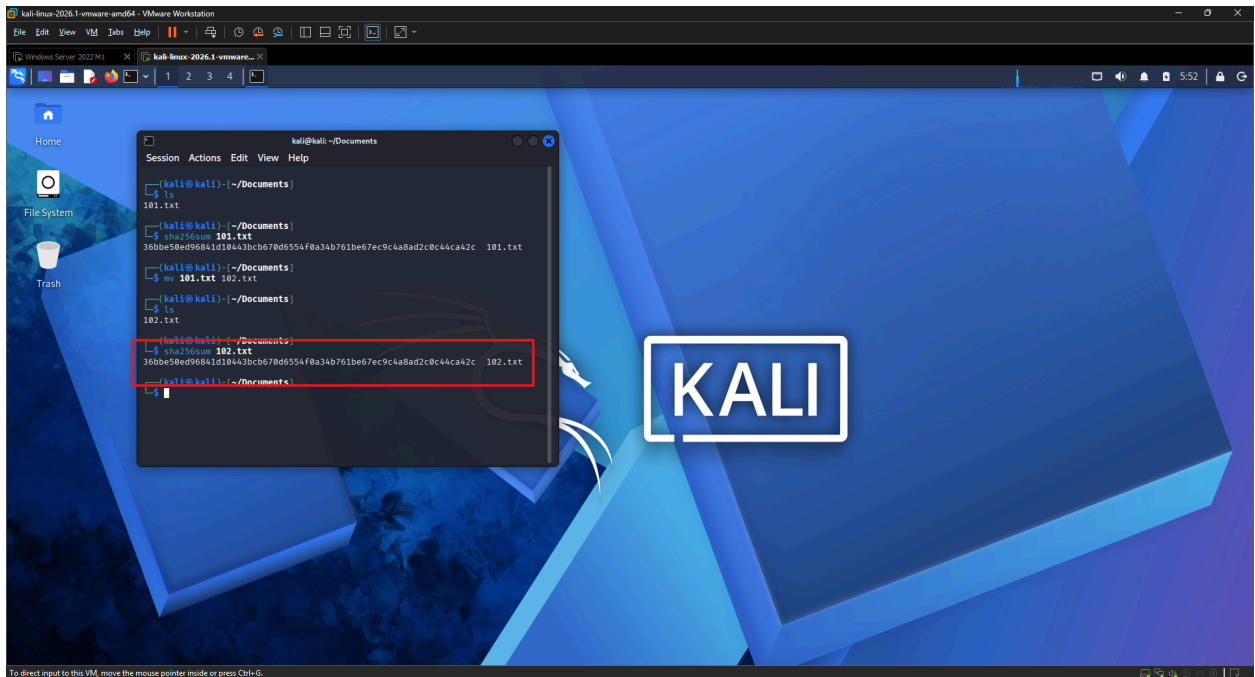


The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal displays the following commands and output:

```
kali@kali: ~/Documents
$ ls
101.txt
kali@kali: ~/Documents
$ sha256sum 101.txt
36bbe58ed9684d10443bcb670d6554f8a34b71be67ec9c4a8ad2c0c44ca42c 101.txt
kali@kali: ~/Documents
$ mv 101.txt 102.txt
kali@kali: ~/Documents
$ ls
102.txt
```

The command ***mv 101.txt 102.txt*** is highlighted with a red box. The terminal window is titled "kali@kali: ~/Documents". The desktop background features a blue and purple geometric pattern with a large "KALI" logo in the center.

2.7 Type the following command: ***sha256sum 102.txt*** to calculate the hash of the renamed file

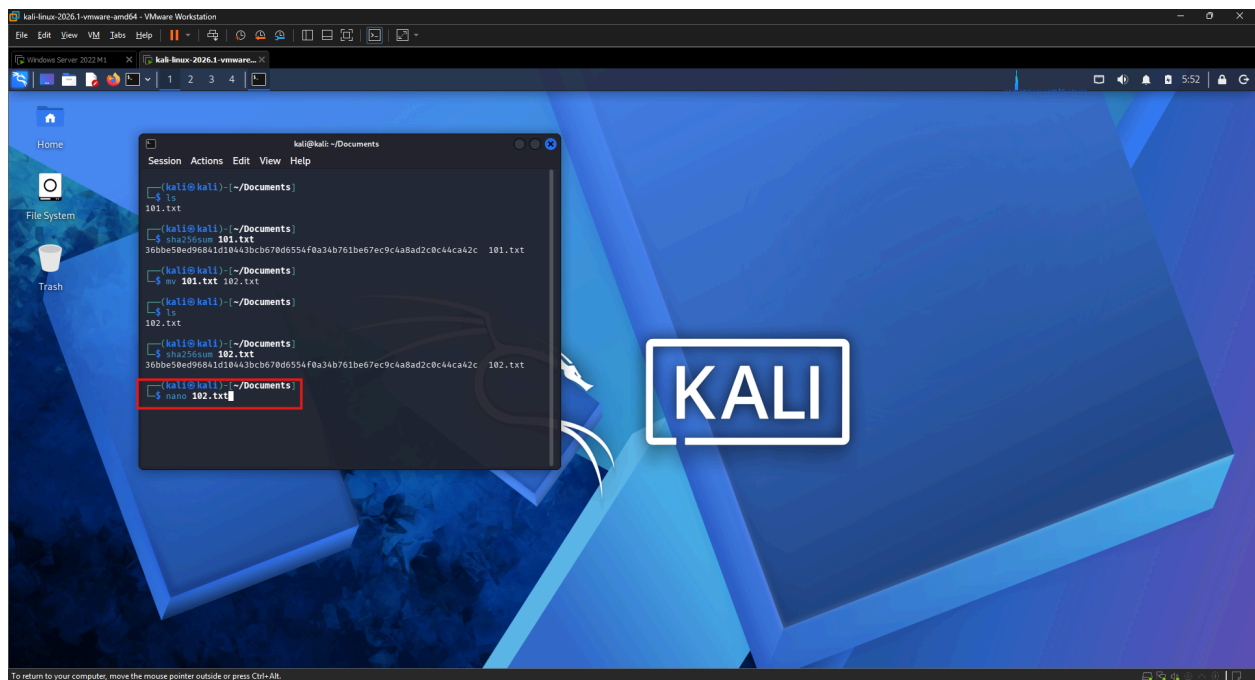


The screenshot shows the same Kali Linux desktop environment with the terminal window open. The terminal displays the following commands and output:

```
kali@kali: ~/Documents
$ ls
101.txt
kali@kali: ~/Documents
$ sha256sum 101.txt
36bbe58ed9684d10443bcb670d6554f8a34b71be67ec9c4a8ad2c0c44ca42c 101.txt
kali@kali: ~/Documents
$ mv 101.txt 102.txt
kali@kali: ~/Documents
$ ls
102.txt
kali@kali: ~/Documents
$ sha256sum 102.txt
36bbe58ed9684d10443bcb670d6554f8a34b71be67ec9c4a8ad2c0c44ca42c 102.txt
kali@kali: ~/Documents
$
```

The command ***sha256sum 102.txt*** is highlighted with a red box. The terminal window is titled "kali@kali: ~/Documents". The desktop background features a blue and purple geometric pattern with a large "KALI" logo in the center.

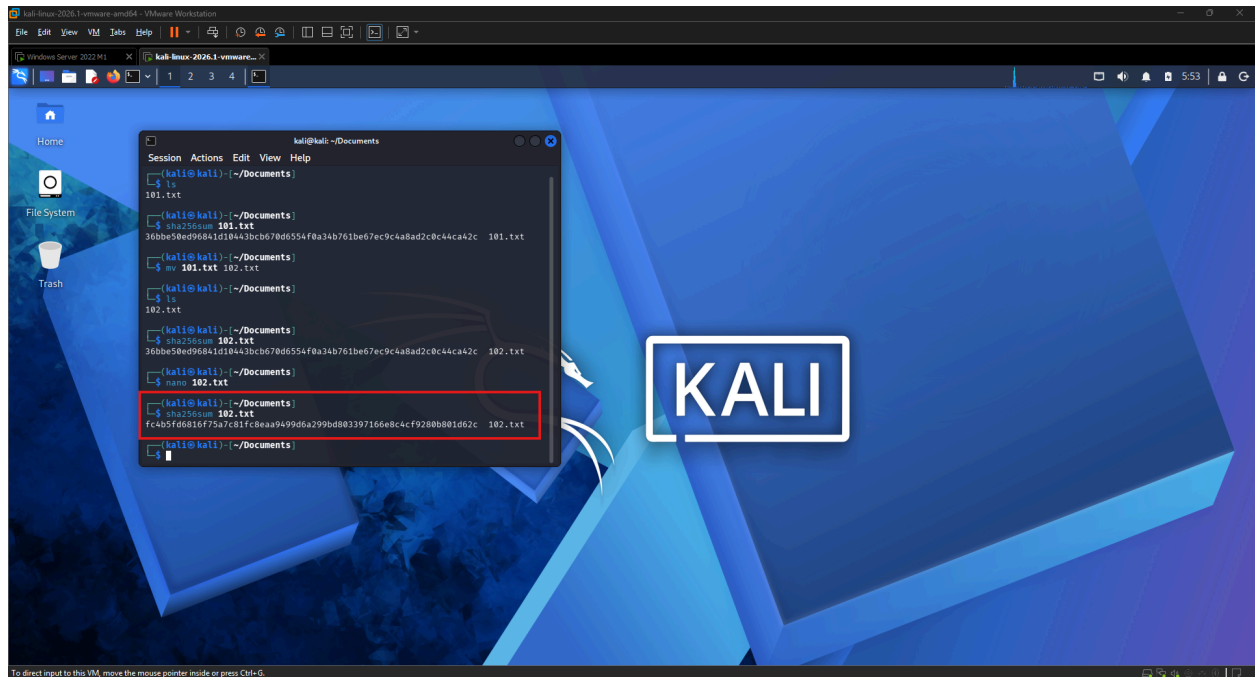
2.8 Type the following command: **nano 102.txt** to open the file for changing its content



2.9 update content and save the file



2.10 Type the following command: ***sha256sum 102.txt*** to find the hash of the modified file



The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal displays the following commands and output:

```
kali@kali: ~/Documents
Session Actions Edit View Help
kali@kali:~/Documents
$ ls
101.txt
$ sha256sum 101.txt
36bbe56ed96841d10443bc670d6554f8a34b761be67ec9c4a8ad2c8c44ca42c 101.txt
$ mv 101.txt 102.txt
kali@kali:~/Documents
$ ls
102.txt
$ sha256sum 102.txt
36bbe56ed96841d10443bc670d6554f8a34b761be67ec9c4a8ad2c8c44ca42c 102.txt
$ nano 102.txt
kali@kali:~/Documents
$ sha256sum 102.txt
fc4b5fd6816f75a7c81fc8aaa9499d6a299bd883397166e8c4cf9280b801d62c 102.txt
$
```

The output of the final `sha256sum 102.txt` command is highlighted with a red box. A large "KALI" logo is visible on the right side of the desktop background.

By following these steps, I have successfully learned how to calculate, transfer, and verify file hashes across different systems, utilizing both Windows Server and Kali Linux environments.